PLATE 01 / 08 · STEP 02 — CUSTOMER OPPORTUNITY

Target Segment *Analysis*

Three segments scored against the Verhaert framework — market value, market access, and our market expertise. Output: a sequenced go-to-market path through the Bulgarian water sector and onward to EU executive agencies.

DATUM	PREPARED	SEGMENTS	CLASSIFICATION
WGS84 · BG	APR 2026	03 / 03 evaluated	INTERNAL

The three segments at a *glance*

Each segment is scored against three dimensions: market value (size, growth, margins), market access (competition, channels, regulatory path), and our market expertise (capital, network, capability). Phosphor pips read 1–5 left-to-right.

SEGMENT 01 · UTILITY

Water Utilities

// Sofiyska Voda · Veolia BG

REACH	1.5 M served
ANNUAL REV.	~€450 M
BUYERS IN BG	~50 ВиК ops
DEAL SIZE	€20–80 K pilot
CYCLE	3–9 mo

MARKET VALUE
MARKET ACCESS
OUR EXPERTISE

VERDICT

Beachhead. Smallest TAM but fastest path to a paying pilot. SV is the lighthouse reference that unlocks segments 2 & 3.

SEGMENT 02 · WMB

Water Mgmt. Boards

// Basin Directorates ×4

COVERAGE	100% BG terr.
RBMP CYCLE	2028–33 (4th)
FUNDING	OPE 21–27 + CF
DEAL SIZE	€80–500 K
CYCLE	9–18 mo

MARKET VALUE
MARKET ACCESS
OUR EXPERTISE

VERDICT

Scalable core. Same product × four geographic customers, EU Cohesion Fund paying. The real prize lives here.

SEGMENT 03 · EXEC

National & EU Agencies

// MOEW · EEA · DG ENV

REACH	27 MS + EU
PROGRAMME	Copernicus
EEA THRESHOLD	€143 K open call
DEAL SIZE	€0.5–5 M FWC
CYCLE	18–36 mo

MARKET VALUE
MARKET ACCESS
OUR EXPERTISE

VERDICT

Long-horizon. Largest TAM but EOMAP, CLS & Brockmann own the channel. Reachable only through 4th RBMP referencing.

STRATEGIC READOUT

Sequence — not selection. Sell SV first, win Danube BD second, reference into the 4th RBMP & EEA call windows third. Skipping segment 1 forfeits the proof point. Skipping segment 2 forfeits the scale.

Segment 01 · *Water Utilities*

Anchor case: Sofiyska Voda, operated by Veolia. Lead persona Elena (Water Quality Manager, Bistritsa LTC). Adjacent BG buyers: ВиК Sofia District, ВиК Plovdiv, ВиК Pernik, ВиК Varna, ВиК Burgas — ~50 regional water-supply operators governed by EWRC.

POPULATION SERVED	SV ANNUAL REVENUE	BG ВИК MARKET	SOURCE-WATER MONITORING SPEND
1.5 M	€450 M+	~€700 M	~1–2%
SV territory · Sofia metro	2025 estimate, RocketReach	all BG operators, EWRC reg.	of OPEX, addressable wedge

① MARKET VALUE

SIZE

SV is the largest WSS operator in BG, serving **~1.5 M** people in Sofia. ~50 regional ВиК utilities in BG aggregate to a sector revenue of **~€700 M**; addressable source-water monitoring slice estimated at **€7–14 M / yr**.

GROWTH RATE

SV revenue trended up from **~€280 M** (2018) toward **~€480 M** (2025) — implied CAGR **~7–8%** driven by tariff approvals and capex pass-through.

PROFIT MARGINS

Bulgarian ВиК operators run thin EBIT margins (regulated, EWRC-capped). NeroSense margin opportunity comes from replacing manual sampling OPEX, not from utility profit share.

DEAL ECONOMICS

Pilot **€20–80 K** (one season, Iskar). Recurring contract **€60–150 K / yr**. LTV anchored to Veolia-Group reference cascade.

② MARKET ACCESS

COMPETITORS

DIRECTLG Sonic (already deployed at SV via aqua3S), CyanoLakes, EOMAP. INDIRECTbuoy makers, manual lab consultancies. SV has no incumbent EO+robot vendor — open lane.

CHANNELS

Direct sale to Elena (Water Quality Mgr) → Operations Director → Veolia Group Digital (Paris). Co-sell with Esri Bulgaria (their ArcGIS partner) shortens the integration story.

REGULATORY PATH

Ordinance No. 9 (drinking water), WHO water-safety plans, reporting to Basin Directorate Pleven & EWRC. No type-approval required for an advisory monitoring layer; ZOP procurement applies above thresholds.

ADOPTION BLOCKER

Vendor-pilot fatigue. SV has seen pilots come and go — close requires a 90-day Iskar pilot with measurable bloom-detection KPI.

③ OUR MARKET EXPERTISE

CAPITAL TO ENTER

Low. Hardware-light deployment uses SV's existing intake-tower buoy + one autonomous surface robot. Pilot cost-of-sale **≤ €30 K**, fundable from CASSINI prize / pre-seed.

NETWORK & CONTACTS

Sofia University „St. Kliment Ohridski" (co-author 2023 Iskar risk study) is our anchor. Warm leads: SV Water Quality team, Sofia Univ. hydrobiology, NIMH. Veolia Paris not yet contacted.

CAPABILITY

Native fit. Sentinel-2/3 fusion + bloom models tuned on Iskar; ArcGIS connector mirrors SV's existing Esri Utility Network. We extend her stack, we don't replace it.

VALIDATION STATUS

Pre-pilot. 2023 SV+Sofia Univ. study explicitly recommends continuing phytoplankton monitoring on raw Iskar water — that recommendation is our scope.

Segment 02 · *Water Management Boards*

The four Bulgarian Basin Directorates (Басейнови дирекции): Danube (Pleven), Black Sea (Varna), East Aegean (Plovdiv), West Aegean (Blagoevgrad). Statutory mandate under Water Act art. 153 implementing WFD 2000/60/EC. Same product profile, four geographic customers.

GEOGRAPHIC COVERAGE	RBMP CYCLE NOW DRAFTING	FUNDING WINDOW	SURFACE WATER BODIES
100%	4th	€2.6 B	~700+
of BG territory · 4 directorates	2028–2033 · methodology open	OPE 2021–27 envelope, BG share	monitored under WFD in BG

① MARKET VALUE

SIZE

Four directorates × monitoring & investigative-study tenders. Realistic addressable value in BG: **€8–15 M** across the 6-year RBMP cycle, expandable to all reservoirs / lakes / significant rivers.

GROWTH RATE

Step-function, not linear. EU Cohesion Fund + OP "Environment" 2021–27 are the lever; the infringement procedure on the 3rd RBMP is forcing accelerated procurement into the 4th cycle.

PROFIT MARGINS

Higher than utility segment. Government contracts price the deliverable, not the marginal cost — EO+robotics service margins of 35–50% are realistic on fixed-scope studies (analogous to the RESAC flood-risk model).

DEAL ECONOMICS

Single-water-body study **€80–250 K**. Multi-year framework **€500 K – 1.5 M**. Per-basin replication rate **×4** if reference deployment lands.

② MARKET ACCESS

COMPETITORS

PRECEDENTRESAC won the Danube BD flood-EO tender — the procurement category already exists. DIRECTBG consultancies (Eko Plus, Balkanika Energy) + NIMH partnerships. EUEOMAP, Brockmann Consult.

CHANNELS

Mandatory ZOP public tender via "Профил на купувача" portal. Procurement favours BG-led consortia. Win pattern: NeroSense + Sofia Univ. + (optional) NIMH as the bidding entity.

REGULATORY PATH

No CE mark required — service deliverable. Methodology must align to WFD Annex V ecological-status protocols. Risk: methodology not yet endorsed in 4th RBMP — must be referenced via Sofia Univ. paper or MOEW pilot before 2028.

ADOPTION BLOCKER

"One-out-all-out" rule. Buyers are politically risk-averse — they need defensible WFD evidence, not better dashboards. Frame value as infringement defense.

③ OUR MARKET EXPERTISE

CAPITAL TO ENTER

Medium. ZOP tenders require performance bonds & financial-capacity proof. We need ~€100–200 K of working capital or a consortium prime to cover bonding (Sofia Univ. or BG consultancy).

NETWORK & CONTACTS

Anchor: Basin Directorate Pleven (already receives SV phytoplankton data). Sofia Univ. — co-author route into MOEW. Cold on Plovdiv, Varna, Blagoevgrad directorates.

CAPABILITY

Strong. Our wedge — phytoplankton coverage on reservoirs + transboundary monitoring (Danube, Struma, Mesta) — maps directly onto WFD biological elements currently underserved.

VALIDATION STATUS

Indirect. SV Iskar pilot + RESAC EO precedent are our two evidence anchors. Need one paid BD study (Danube, ideally) before scaling to the other three.

Segment 03 · *National & EU Executive Agencies*

Two layers. National: MOEW Water Management Directorate, Executive Environment Agency (ExEA / ИАОС). European: EEA (Copenhagen), JRC (Ispra), DG ENV, EUMETSAT, ECMWF — entrusted entities of the Copernicus programme. Buyers of methodology, framework contracts, and continental data products.

COPERNICUS ENVELOPE	EEA TENDER THRESHOLD	MEMBER STATES SERVED	WFD ECOLOGICAL STATUS
€5.4 B	€143 K	27	37%
2021–27 EU Space Programme	open-call procedure floor	+ Norway, Iceland, UK	EU surface waters at "good" status

① MARKET VALUE

SIZE

Single largest TAM. Copernicus 2021–27 envelope **~€5.4 B**; EEA Land Monitoring & In-Situ component framework contracts represent the addressable downstream market. WFD-related EEA & JRC procurement: **€20–40 M / yr** across the EU.

GROWTH RATE

Strong. Driven by Green Deal monitoring obligations, Nature Restoration Law, the recast Drinking Water Directive, and the upcoming Urban Wastewater Treatment Directive — all need EO-derived evidence at MS scale.

PROFIT MARGINS

Highest of the three. Multi-year framework service contracts (FWC) carry margins comparable to ESA prime contractors — ~30–45% on services. Bid-cost recovery is the bottleneck, not unit economics.

DEAL ECONOMICS

EEA call for tenders typically **€500 K – 5 M** per lot; framework contracts run 2–4 years renewable. JRC FWCs (e.g. CEMS Hydrological Forecast Centre) are larger but require Tier-1 prime status.

② MARKET ACCESS

COMPETITORS

DOMINANTEOMAP (DE) — UNESCO IIWQ portal, ESA BIC alumnus. DOMINANTBrockmann Consult (DE), CLS (FR), GMV (ES). EMERGINGSinergise (BlueDot), Gybe (US), CyanoLakes (ZA). Crowded but specialised.

CHANNELS

EU TED portal & EEA Funding-and-Tender portal for >**€143 K** calls. Direct procurement under threshold via negotiated procedure. Most contracts won by consortia of 3–5 entities.

REGULATORY PATH

No product certification — service quality validated against CMEMS / CLMS reference protocols & published peer-reviewed methodology. Pre-qualification: ISO 9001, financial-capacity check (~3× annual contract value).

ADOPTION BLOCKER

Three-fold. (a) Reference customers — must show MS deployments first. (b) Bid cost of €30–80 K per major tender. (c) Methodology must be cited in WFD/CIS guidance to be considered "operational-grade".

③ OUR MARKET EXPERTISE

CAPITAL TO ENTER

High barrier. Bid costs, financial guarantees, and 12–24 month payment cycles require either Series-A funding or partnership with an EU prime (Telespazio, e-GEOS, Spacebel) carrying us as a sub.

NETWORK & CONTACTS

Cold. No direct contact with EEA Copernicus team, JRC Ispra, or DG ENV. Path-in: CASSINI mentorship, Copernicus Masters, ESA BIC, and Sofia Univ.'s EurAqua membership.

CAPABILITY

Technically credible. Our use of Sentinel-2/3 + Modular Inversion is on-par with EOMAP's approach. The differentiator is the EO + autonomous in-situ closed loop — that combination is not in EOMAP's stack.

VALIDATION STATUS

Not yet operational at MS scale. Requires segments 1 & 2 to deliver references first. Realistic first EEA bid is 2027–2028, after BG track record is established.

Comparison *matrix*

Verhaert "Choose the target segment" framework, fully populated. Read vertically per dimension; read horizontally per segment. Tags: ● favourable · ● caution · ● blocker.

DIMENSION	SEGMENT 01 — WATER UTILITIES Sofiyska Voda · ВиК operators	SEGMENT 02 — WATER MGMT. BOARDS Basin Directorates ×4	SEGMENT 03 — NATIONAL & EU AGENCIES MOEW · EEA · JRC · DG ENV
MARKET VALUE			
SIZE Addressable monitoring spend	BG ВиК sector **~€700 M** rev. · monitoring slice **€7–14 M / yr**.	BG basin monitoring tenders **€8–15 M / 6-yr cycle**.	EU EEA + JRC water-EO procurement **€20–40 M / yr**.
GROWTH RATE 5-year outlook	+7% CAGR tariff-driven, EWRC-regulated.	STEP-UP 4th RBMP unlock 2028, infringement-accelerated.	STRONG Green Deal + Nature Restoration + DWD recast.
PROFIT MARGINS Service / product margin available	**Tight**. Utility OPEX squeeze; we win on cost-savings, not premium pricing.	**35–50%** on fixed-scope studies. RESAC precedent supports.	**30–45%** on multi-year FWC, but bid-cost recovery is the gate.
MARKET ACCESS			
COMPETITORS Who already plays here	LG Sonic (in-situ ultrasonic, deployed at SV), CyanoLakes, EOMAP. OPEN LANE for EO+robotics combo.	RESAC (flood-EO precedent), BG consultancies, NIMH-led bids. CROWDED LOCAL .	EOMAP, Brockmann, CLS, GMV, Sinergise. CROWDED EU .
CHANNELS & DISTRIBUTORS How to reach the buyer	Direct sale + Esri Bulgaria co-sell + Veolia Group Digital path to scale.	ZOP tenders via "Профил на купувача"; **BG-led consortium required**.	EU TED portal, EEA Funding-and-Tender portal, sub-contractor to EU primes.
REGULATORY PATH Compliance & approvals	NO TYPE-APPROVAL . Ordinance No. 9, EWRC reporting fits with our audit trail.	Methodology must align to WFD ANNEX V ; aim for 4th RBMP reference by 2028.	ISO 9001 + financial capacity + WFD/CIS guidance citation. LONG PRE-QUAL .
OUR MARKET EXPERTISE			
FINANCIAL RESOURCES TO ENTER Cost-of-sale & runway	**≤€30 K** pilot CoS. WITHIN REACH on CASSINI prize / pre-seed.	**€100–200 K** working capital + bonds. NEEDS CONSORTIUM PRIME .	**€30–80 K** per major bid + 12–24 mo payment cycle. NEEDS SERIES-A OR SUB-CONTRACTING .
NETWORK & CONTACTS Warm leads in segment	STRONG . SV Water Quality team, Sofia Univ. hydrobiology, NIMH.	PARTIAL . Basin Dir. Pleven warm; other 3 BDs cold.	COLD . No EEA/JRC/DG ENV contacts. CASSINI & Copernicus Masters are the entry vector.
CAPABILITY & MARKET KNOWLEDGE Product–segment fit	NATIVE FIT . EO+robot extends Elena's existing buoy + ArcGIS stack.	STRONG . Phytoplankton + transboundary monitoring map to WFD biological elements.	CREDIBLE . EO+in-situ closed loop is differentiated, but operational maturity is missing.
COMPOSITE			
SCORE (× / 15) 3 dim. × 5-pt scale	**12 / 15** · BEACHHEAD	**11 / 15** · SCALE	**9 / 15** · HORIZON
TIME-TO-REVENUE First paid contract	**3–9 months**	**9–18 months**	**18–36 months**

Strategic *recommendation*

The three segments are not alternatives — they are a sequence. Win Segment 01 to build the reference. Use that reference to win Segment 02 at scale. Use both to qualify into Segment 03. The mistake to avoid: chasing Segment 03's TAM before earning the references that gate entry.

01
PRIMARY · BEACHHEAD

Segment 01 — Water Utilities

Sofiyska Voda is the only segment where we can land a paying contract within the next two quarters. The product is already shaped for Elena's stack; our network here is the strongest; the regulatory path requires no certification.

The pilot is not just revenue — it is the **reference deployment** that every later customer will ask to see. The Iskar pilot is therefore strategic infrastructure for segments 2 and 3, not a standalone bet.

NEXT 90 DAYS

- Validation interview with Elena + Operations Director (Bistritsa LTC).
- Co-author one-pager methodology with Sofia University.
- Scope a 90-day Iskar pilot, deliverable-priced **<€30 K**.
- Loop in Esri Bulgaria as integration co-seller.

02
CONCURRENT · SCALE

Segment 02 — Basin Directorates

Begin warm-up while the Iskar pilot runs. Basin Directorate Pleven already receives SV phytoplankton data — **they are downstream of the pilot we are about to deliver**. The infringement procedure on the 3rd RBMP and the open methodology window for 2028–33 mean the procurement frame is being written this year.

The asymmetric outcome: getting a Sofia-University-co-authored methodology referenced (even informally) into the 4th RBMP locks in 6 years of procurement across all four basins.

NEXT 6–12 MONTHS

- Validation interviews with Senior Monitoring Expert + Director "Plans & Permits" at Pleven.
- Identify a BG consortium prime (Sofia Univ. + NIMH preferred).
- Tender-watch the "Профил на купувача" pages of all four BDs.
- Submit one ZOP bid — Pleven, single water body — by Q4.

03
DEFERRED · HORIZON

Segment 03 — Executive Agencies

Largest TAM, but EOMAP, Brockmann, CLS and GMV own the EEA channel today. We do not have the references, the financial capacity, or the EU primes' rolodex to bid the major framework contracts — yet.

The right play is **channel-building, not bidding**. CASSINI mentorship, Copernicus Masters submission, and a paper co-authored with Sofia Univ. are the cheapest, highest-leverage moves. First EEA / DG ENV bid realistic in 2027–28.

NEXT 12–24 MONTHS

- CASSINI Hackathon win → ESA BIC application path.
- Submit Copernicus Masters with Iskar use-case.
- Co-author one peer-reviewed paper on EO+in-situ closed loop.
- Identify two EU primes for sub-contractor positioning.

×
OUT OF SCOPE (FOR NOW)

What we are *not* doing

Industrial water users (mining tailings, brewing, aquaculture) are real customers but require a different sales motion and case-mix. Park until 2027.

Hardware-only sales of the autonomous surface robot. The robot is part of the closed loop, not a stand-alone SKU. Selling it separately devalues the platform thesis.

Direct B2C / municipal-citizen dashboards. Public information is a **freebie bundled** into a paid BD contract, never a primary deliverable.

RE-EVALUATE

- Q4 2027 — after first BD contract delivers.

Sequenced *roadmap*

Timeline reads left-to-right; segments are colour-tagged. Crossover points are where one segment's deliverable becomes the next segment's evidence.

PHASED ROLLOUT · 2026 → 2029

Q2 2026	SEG 01 Validation interviews with Elena (SV Water Quality) + Sofia University co-author scoping. CASSINI submission.
Q3 2026	SEG 01 Iskar 90-day pilot kicks off. Buoy + autonomous surface robot deployed. KPI: bloom-detection lead-time vs intake-buoy baseline.
Q4 2026	SEG 01 Pilot results report. SEG 02 Warm-up at Basin Dir. Pleven. Tender-watch begins.
Q1–Q2 2027	SEG 01 SV recurring contract signed (~€80 K/yr). Veolia Paris briefed. SEG 03 Copernicus Masters submission with Iskar use-case.
Q3 2027	SEG 02 First ZOP bid submitted at Danube BD (single water body, ~€150 K). BG consortium with Sofia Univ. as scientific lead.
Q4 2027 – Q1 2028	SEG 02 First BD study delivered. Methodology proposed for inclusion in the 4th RBMP draft.
Q2 2028	SEG 02 Replication bids prepared for Black Sea, East Aegean, West Aegean BDs. SEG 03 First EEA tender bid as sub-contractor to an EU prime.
2029	SEG 02 Multi-basin framework contract win. SEG 03 Lead bid on a small EEA Land Monitoring lot (water-quality component).

KEY ASSUMPTIONS

A1 — ISKAR PILOT	Pilot returns measurable bloom-detection lead-time improvement vs the intake-tower buoy baseline. Failure here halts segments 2 & 3.
A2 — RBMP CYCLE	4th cycle methodology window stays open through 2027. EU infringement on 3rd cycle is not retroactively quashed.
A3 — FUNDING	OP "Environment" 2021–27 disbursement timetable holds. Cohesion Fund flow into BG basins continues at projected pace.
A4 — CONSORTIUM	Sofia University formally agrees to co-author methodology & serve as scientific prime for ZOP bids.

RISKS & MITIGATIONS

R1 — VEOLIA VETO	Paris HQ requires global vendor approval. Mitigation: pitch SV as Veolia-Group reference deployment, not a one-off.
R2 — TENDER LOSS	First Danube BD bid lost to incumbent BG consultancy. Mitigation: bid as sub-contractor to a BG prime first, then prime second tender.
R3 — METHODOLOGY GAP	4th RBMP closes without referencing EO methods. Mitigation: publish peer-reviewed paper before 2027 for citation cover.
R4 — COMPETITIVE ENTRY	EOMAP or CyanoLakes opens a Sofia office. Mitigation: speed of pilot + BG-led consortium are our moat — they cannot replicate the local network quickly.